# Accenture — Aptitude Test 3

Questions 51–75 of 100 · 25 questions

**Question 51. In an examination involving quantitative aptitude and logical reasoning, 65% examinees cleared quantitative aptitude test while 70% cleared logical reasoning test. If 50% examinees passed both the tests, then how many failed in both tests?**

**Question 52. Nitish sold his watch and sunglasses at a loss of 4% and gain of 4% respectively for 2600 to Kamal. Kamal sold the same sun glasses and watch at a loss of 4% and gain of 4% respectively for 2700. The price of watch and sun glasses to Nitish were?**

A c c e n t u r e

If initial value (I.V) is 100, then the final value(F.V) will be 180
180 = (6/5)(5/4)[(100+c)/100] x 100
180 = (30/20)(100 + c)
100 + c = 180 x (20/30) = 120
c = 20
The third profit percentage is 20%

Solution:
Assume that the cost price of watch is A.

100+a

100+a

100
x
x I.V

100

100

x 100+c

120

x

100

100

**Question 53. If the price of gold increases by 30%, find by how much the quantity of ornaments must be reduced so that the expenditure may remain the same as before?**

**Question 54. The price of a jewel, after passing through three merchants rises on the whole by 80%. If the first and the second merchants earned 20% and 25% profit respectively, find the percentage profit earned by the third merchant.**

Solution:
Percentage passed in Quant alone
= 65 – 50 = 15%
Percentage passed in reasoning alone
= 70 – 50 = 20%
Percentage passed in at least one subject
= 15 + 20 + 50 = 85%
Percentage failed in both=100–85=15%

Solution:
Formula to find final value after 3 successive profits is:

x
F.V =
100+a

100
F.V =
x
I.V
125

**Question 55. An exhibition was conducted for 4 weeks. The number of tickets sold in 2nd work-week was increased by 20% and increased by 16% in the 3rd workweek**

**but decreased by 20% in the 4th workweek. Find the number of tickets sold in the beginning, if 1392 tickets were sold in the last week.**

Solution:
Tickets in 4th week = (80/100) 3rd week
1392 = (80/100)3rd week
3rd week = 1392 x 100/80 = 1740
3rd week = (116/100) 2nd week
2nd week = 1740(100/116) = 1500
2nd week = (120/100)1st week
1st week = 1500 x (100/120) = 1250.
Number of tickets in 1st week = 1250

**Question 56. Consider three brothers Ram, Ravi and Rahul. Consider Ram to be taller than Ravi by 10% and Rahul is taller than Ravi by 30%. Now, by how much percentage Rahul is taller than Ram?**

Solution:
Assume that the height of Ravi = 100cm.
Height of Ram = 110cm.
Height of Rahul = 130cm.
Percentage increase from height of Ram to Rahul:
% = [(130 – 110)/110] x 100 = 18.18%

**Question 57. Seats for Mathematics, Physics and Biology in a school are in the ratio 5 : 7 : 8. There is a proposal to increase these seats by 40%, 50% and 75% respectively. What will be the ratio of increased seats?**

Solution:
Assume that the number of students in each department is 50, 70 and 80.
After increasing,
Maths = 50 + (40/100) x 50 = 70
Physics = 70 + (50/100) x 70 = 105

Biology = 80 + (75/100) x 80 = 140
Number of students after increasing
= 70, 105, 140
Ratio = 2 : 3 : 4

**Question 58. What will Rs.1500 amount to in three years if it is invested in 20% p.a. compound interest, interest being compounded annually?**

Solution:
Amount
= (120/100)(120/100)(120/100)1500
= (6/5)(6/5)(120)(15)
= 6 x 6 x 24 x 3
= Rs. 2592

**Question 59. The number of degrees that the hour hand of a clock moves through between noon and 2.30 in the afternoon of the same day is?**

Solution:
Hour hand moves 30 degrees in 1 hour.
Time between 12:00 and 2:30 = 2.5hr
Degrees made by hour hand
= 2.5 x 30 = 75 degrees.

**Question 60. A clock is started at noon. By 10 minutes past 5, the hour hand has turned through how many degrees?**

Solution:
Time taken from noon to 10 minutes
past 5 = 5 hours and 10 minutes
= 5.166 hours.
Degrees made by hour hand
= 5.166 x 30 = 155 degrees.

**Question 61. My husband's watch gains 2 minutes every hour and my watch loses 1 minute for each hour. One day, we were late to marriage because the difference between the time in the two watches was 1 hr. and we looked at the slow watch. When did we last set our watches to the same time?**

A c c e n t u r e

Solution:
The delay of 1 hour is because husband was 40 minutes early and the wife was 20 minutes late.
(Since the ratio between error time is 2:1). According to the wife's watch they have set the time 20 hours before.

Un-grazed area = Area of square – 4 Area of quarter circle
= 142 – (22/7)72

## Question 62. A person travels through 5 cities - A, B, C, D, E. City E is 2 km west of D. D is 3 km north-east of A. C is 5 km north of B and 4 km west of A. If this person visits these cities in the sequence B - C - A - E - D, what is the effective distance between cities B and D?

= 196 – 154 = 42.
Grazed area = 154 m2

It takes 11 days to graze 154 m2.
Time taken to graze 42 m2 will be:
154/42 = 11/N
N = 11 x 42/154 = 3 days.

Solution:
The question can be represented in a diagram as follows.

## Question 63. Four horses are there at the four corners of a square of side 14m such that two horses along the same side can just reach each other. They were able to graze the area in 11 days. How many days will they take in order to graze the left out area?

## Question 64. If the diagonal and the area of a rectangle are 25 m and 168 m2, what is the length of the rectangle?

D
E
O

Solution:
Let diagonal be D, which is 25.
D2 = L2 + B2

C
A

625 = L2 + B2

L x B = 168
(L + B)2 = L2 + B2 + 2LxB
(L + B)2 = 625 + 2 x 168
(L + B)2 = 961 ■L + B = 31
L + B = 31 and L x B = 168
By solving we get, L = 24 and B = 7

B

OA = OD = side of the square with
diagonal AD.
Diagonal AD = 3.
Side OD = 3/√2 = 2.12
OE = OD – ED = 0.12
AE2 = OA2 + OE2

## Question 65. 10 men can complete a piece of work in 15 days and 15 women can complete the same work in 12 days. If all the 10 men and 15 women work together, in how many days will the work get completed?

AE2 = 2.122 + 0.122

AE ■1.5
Effective distance for B-C-A-E-D
= 5 + 4 + 1.5 + 2 = 12.5km.

Solution:
Time taken by two groups together
= (AB)/(A+B) = (15 x 12)/(15 + 12)
= 180/27 = 6.67 days.

## Question 66. A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?

Solution:
Shaded part is the area grazed by the
horse when tied to the corners.

A c c e n t u r e

Solution:
Work done by A in 1 day = 1/20
Work done by B in 1 day = 1/30
Work done by C in 1 day = 1/60
Work done in 1st and 2nd day =2(1/20)
= 1/10
Work done on the third day
= (1/20) + (1/30) + (1/60) = 1/10

Work done in three days
= (1/10) + (1/10) = 1/5
Time taken to complete the work will be
equal to 5 x 3 = 15 days

**Question 67. If 20 men or 24 women or 40 boys can do a job in 12 days working for 8 hours a day, how many men working with 6 women and 2 boys take to do a job four times as big working for 5 hours a day for 12 days?**

Solution:
24 women = 20 men; 6 women = 5 men
40 boys = 20 men; 2 boys = 1 man
Le the number of men required be N.
So, N + 5 + 1 = N + 6 men are there in the second case.

W1R2T2H2 = W2R1T1H1
1 x (N + 6) x 12 x 5 = 4 x 20 x 12 x 8
(N + 6) = 128
N = 122.
122 men are required.

**Question 68. Each helper can make either 2 large cakes or 35 small cakes per hour. The kitchen is available for 3 hours and 20 large cakes and 700 small cakes are needed. How many helpers are required?**

Solution:
1 helper can make 2 large cakes in 1st

hour and 70 small cakes in the next 2 hours.
If there are 10 helpers, they can make 20 large cakes in first hour and 700 small cakes in next 2 hours.
Total helpers required = 10.

**Question 69. A fort has enough food for 45 days for 175 soldiers. If after 15 days 100 soldiers leave the fort, for how many more days the food will last?**

Solution:
After 15 days there will be food for 175 soldiers for 30 days.
The number of soldiers for 30 days = 75
(30 x 175) = D x 75■D = 70 days.

**Question 70. A can do a piece of work in 36 days, B in 54 days and C in 72 days. All of them began together but A left 8 days and B**

**left 12 days before the completion of the work. How many days in all did C put in till the entire work was finished?**

Solution:
Let the number of days worked by C = N
Number of days worked by A = N – 8
Number of days worked by B = N – 12
Work done by all A, B and C:
[(N–8)/36]+[(N–12)/54]+N/72 = 1
6(N – 8) + 4(N – 12) + 3N = 216
6N – 48 + 4N – 48 + 3N = 216
13N = 312
N = 24. C worked for 24 days.

**Question 71. One man or two women or three boys can do a work in 44 days. How many days will one man, one women and one boy together take to complete the work?**

Solution:
2 women= 1 man; 1 woman= (1/2)man
3 boys = 1 man; 1 boy = (1/3) man.
1 man + 1 woman + 1 boy
= 1 + (1/2) + (1/3) men = 11/6 men.
If 1 man can complete the work in 44 days and 11/6 men can do it in N days, then,
[1/(11/6)] = N/44
N = 24
They can complete the job in 24 days.

**Question 72. If a certain computer is capable of printing 4900 monthly credit card bills per hour, while a new model is capable of printing at a rate of 6600 per hour, the old model will take approximately how much longer than the new model to print 10000 bills?**

A c c e n t u r e

Solution:
Time taken by old model = 10000/4900
Time taken by new model= 10000/6600
Difference= 10000/4900 – 10000/6600
= 0.5253 hours
= 31 minutes 32 seconds

**Question 73. Three friends Gerald, Rooney and Ronaldo work together to dig a hole. Gerald alone can complete the work in 10 days, Ronaldo in 8 days and together**

**all three can complete it in 4 days. They earn a total of Rs.1,200. Find the share of Rooney if the money that they received is proportional to the work that they do?**

Solution:
While A completes the race, B is 6 meters ahead of C.
When B completes the race, B is 8 meters ahead of C.
B takes 12 meters to get a lead of 2 m.
So, for 8m lead he needs 4 x 12 = 48m.
Length of the rave is 48 meters.

Solution:
Let the time taken by Gerald, Rooney and Ronaldo to complete the work be A, B and C respectively.
A = 10; C = 8
Given, (1/A) + (1/B) + (1/C) = ¼
(1/10) + (1/B) + (1/8) = ¼
By solving, we get B = 40
Ratio of their salary is proportional to ratio of their 1 day work.
Salary ratio of:
Gerald : Rooney : Ronaldo
= (1/10) : (1/40) : (1/8)
= 4 : 1 : 5
Share of Rooney = (1/10) x 1200 = 120

**Question 74. Two pipes can fill a tank in 10 hours and 12 hours respectively while a third, pipe empties the full tank in 20 hours. If all the three pipes operate simultaneously, in how much time will the tank be filled?**

2 – 4)]
11Sb

2 – 44 = 240Sb
11Sb

2 – 240Sb – 44 = 0
By solving this, we get
Sb = 22
Speed of boat in still water = 22kmph

Solution:
Let T be the time taken together.
(1/T) = (1/10) + (1/12) – (1/20)
(1/T) = (6/60) + (5/60) – (3/60)
(1/T) = 8/60 ■T = 60/8 = 7.5 hours

**Question 75. A, B and C can do a work in 5 days, 10 days and 15 days respectively. They started together to do the work but after 2 days A and B left. How long did C take to finish the remaining work?**

A c c e n t u r e

Solution:
Distance = Time x relative speed
Distance between Riya and Priya after 30 minutes
= (30/60) x (20 + 30) = 0.5 x 50
(since opposite direction add the speed)
= 25 km.

Solution:
Work done in 1st two days
= 2[(1/5) + (1/10) + (1/15)] = 22/30
Remaining work = 8/30
Time taken for C alone:
= 15 x (8/30) = 4 days.